

NETWORK MODELS — WHITEBOARD PRESENTATION CHEAT SHEET

Erdős–Rényi · Watts–Strogatz · Barabási–Albert

TIMING — KEEP THIS IN VIEW

Time	Checkpoint	Section
0:00–0:45		Opening — one sentence + name 3 models
0:45–3:30	✓ ER done	Erdős–Rényi $G(n,p)$
3:30–6:00	✓ WS done	Watts–Strogatz
6:00–8:30	✓ BA done	Barabási–Albert
8:30–9:30		Comparison table
9:30–10:00		Closing sentence + questions

⚠ If ER is not done by 3:30, cut a bullet point — NEVER cut the comparison table.

OPENING (0:00–0:45)

Say aloud (nothing on the board yet):

"Three models, three different answers to the same question: how do real networks form?"

Then write this list in the top-left corner — keep it there ALL presentation:

1. Erdős-Rényi $G(n, p)$
2. Watts-Strogatz
3. Barabási-Albert

These 3 lines take ~20 seconds to write. Never erase them.

MODEL 1: Erdős-Rényi $G(n, p)$ (0:45–3:30)

Write on Board

$G(n, p)$: n vertices, each edge included independently with prob. p
 $E[|E|] = p \cdot C(n, 2)$
 $d(v) \sim \text{Bin}(n-1, p), \quad E[d(v)] = p(n-1)$
Connectivity threshold: $p_c = (\ln n) / n$

Draw on Board

- Six dots in a rough circle
- Draw 4–5 random edges between some pairs — label a couple 'prob. p '
- Leave some pairs unconnected — independence is the point
- No need to label individual vertices

Say (memorise these points)

- Each edge exists or not, independently — no memory, no preference
- Degree is Binomial: each of the $n-1$ other vertices contributes an independent Bernoulli
- Below threshold p_c : isolated vertices almost surely survive
- Above threshold: graph is almost surely connected
- At $p = 1/n$: a giant component of size $\Theta(n)$ suddenly appears — the phase transition
- Degree distribution is tightly concentrated around the mean — no heavy tails
- Clustering coefficient $\approx p$, which is small for sparse graphs — no local structure

💡 Key Insight: ER gives us a sharp threshold for connectivity, but produces homogeneous, unstructured graphs — nothing like a real social network.

Transition line:

"So we ask: can we build in local clustering while keeping short path lengths?"

MODEL 2: Watts–Strogatz (3:30–6:00)

Write on Board

Construction:

1. Ring lattice: n vertices, each connected to k nearest neighbours
2. Rewire each edge independently with prob. $\beta \in [0,1]$

$\beta = 0 \rightarrow$ lattice: high clustering, large diameter

$\beta = 1 \rightarrow$ random: low clustering, short paths

β small $\rightarrow$ small-world: HIGH clustering + SHORT paths $\leftarrow$ sweet spot

Draw on Board

- A ring of ~ 8 dots, each connected to 2 nearest neighbours each side $\rightarrow$ label ' $k=4$ '
- Draw 1–2 edges crossing the ring $\rightarrow$ label 'rewired (prob. β)'
- Annotate: 'shortcuts reduce diameter dramatically'
 - One diagram is enough — ring + a couple of crossing edges

Say (memorise these points)

- Start with structure: a lattice has order but is inefficient to traverse
- A few random shortcuts collapse the diameter — 'six degrees of separation'
- Clustering stays high because most edges are untouched, triangles preserved
- Average path length drops from $O(n/k)$ on lattice $\rightarrow O(\log n)$ with a handful of shortcuts
- β is the single control parameter — interpolates between two extremes
- Small-world regime is not one value of β but a wide intermediate range
- Captures social network and neural network structure well

💡 Key Insight: High clustering and short path lengths are not opposites — WS shows they coexist with just a few random shortcuts.

Transition line:

"But neither model explains why real networks have a tiny number of extremely high-degree nodes — hubs. For that, we need a growth mechanism."

MODEL 3: Barabási–Albert (6:00–8:30)

Write on Board

Growth: add one new vertex + m edges at each time step

Preferential attachment:

$$P(\text{connect to vertex } i) = d_i / \sum_j d_j$$

Degree distribution: $P(k) \sim k^{-3}$ (power law)
→ Scale-free network

Draw on Board

- A star graph: one central hub with 5–6 leaves → label hub 'high degree'
- Add a new dot at the edge, draw arrow to the hub → label 'new node attaches here with high prob.'
- Annotate: 'rich get richer'

★ This is your most important diagram — make it large and clear.

Say (memorise these points)

- The model is DYNAMIC — graph grows over time, unlike ER and WS
- New nodes are more likely to connect to already-popular nodes — positive feedback
- Preferential attachment: probability proportional to current degree
- Power law $P(k) \sim k^{-3}$ has a heavy tail: most nodes low degree, a few enormous
- 'Scale-free' = no characteristic scale; distribution looks the same at any zoom
- Hubs are both strength (explains real networks) and vulnerability (targeted attack removes them)
- Exponent -3 is derived via mean-field analysis — it is a prediction, not a fit

💡 Key Insight: Growth plus preferential attachment is sufficient to generate the heavy-tailed degree distributions we observe in the web graph, citation networks, and protein interaction networks.

COMPARISON SUMMARY (8:30–9:30)

Draw this table on the board — write quickly, columns already in your head:

	Erdős–Rényi	Watts–Strogatz	Barabási–Albert
Mechanism	Random edge placement	Rewiring lattice edges	Growth + pref. attachment
Degree dist.	Binomial $\approx$ Poisson	$\sim$ Poisson (narrow)	Power law k^{-3} (heavy tail)
Clustering	Low ($\approx p$)	HIGH	Low–moderate
Hubs	No	No	YES
Path length	Short ($O(\log n)$)	Short ($O(\log n)$)	Short ($O(\log n / \log \log n)$)
Real-world fit	Threshold / connectivity	Social & neural nets	Web, citation, protein nets

Say: "Each model isolates one mechanism. Together they explain complementary features of real networks."

Closing (9:30–10:00)

"Real networks are not random, not purely local, and not static."

"ER gives us structure and thresholds. WS gives us locality and efficiency. BA gives us hubs and history."

"Together, they form a toolkit for thinking about any networked system."

"Happy to take any questions."

BOARD MANAGEMENT RULES

- TOP-LEFT: Permanent model list — never erase it
- Work left to right: ER on left third · WS in middle · BA on right
- Only erase if you genuinely run out of space — never erase model list
- Write formulas large enough to read from 3 metres
- Leave whitespace — sparse looks confident, crowded looks panicked

NEVER say:

- 'In my code...' or 'When I ran the simulation...'
- Anything about Python, simulations, or computational results
- 'Sorry about my handwriting' — do not apologise for the board
- Do NOT read directly from what you wrote — write it, then turn and explain it

LIKELY QUESTIONS & PREPARED ANSWERS

Question	Key Point to Make
What happens at exactly $p = p_c$?	Giant component emerges. Below: only small components of size $O(\log n)$. Above: almost surely connected.
Why is the BA exponent exactly -3 ?	It follows from the mean-field rate equation for degree growth — a theoretical prediction, not a curve fit.
Can WS produce a power-law distribution?	No. Degree distribution in WS is approximately Poisson for small β . No heavy tails.
How does BA handle clustering?	It doesn't — BA clustering is low. Hybrid models like Holme–Kim add triangle-closing steps on top of BA.
What is the clustering coefficient formally?	$C(v) = (\text{triangles through } v) / C(\text{degree}(v), 2)$. Ratio of actual to maximum possible triangles.
Are these models realistic?	Each captures one feature. Real networks usually need hybrid or more complex generative models.
What is the phase transition in ER?	At $p = 1/n$ a giant component of size $\Theta(n)$ suddenly appears. Below: fragmented. Above: one dominant component.
How is WS different from pure random?	WS preserves local structure (triangles) while adding shortcuts. ER has neither clustering nor locality.

FORMULA QUICK REFERENCE

Erdős–Rényi

$G(n, p)$ — n vertices, edges i.i.d. Bernoulli(p)
 $E[|E|] = p \cdot C(n, 2) = p \cdot n(n-1)/2$
 $d(v) \sim \text{Bin}(n-1, p)$ $E[d] = p(n-1)$
 Phase transition: $p_c = \ln(n) / n$
 At $p = 1/n$: giant component $\Theta(n)$ appears
 Clustering $\sim p$ (small for sparse graphs)

Watts–Strogatz

Ring lattice: n nodes, each connected to k nearest neighbours
 Rewire each edge with prob. $\beta \in [0, 1]$
 $\beta = 0$: $C \approx 3(k-2)/4(k-1)$, $L = O(n/k)$
 β small: C still $\approx 3(k-2)/4(k-1)$, $L = O(\log n)$
 $\beta = 1$: $C \approx k/n$, $L = O(\log n)$

Barabási–Albert

At each step: add 1 node + m edges
 $P(\text{attach to } i) = d_i / \sum_j d_j$
 Steady-state: $P(k) \sim k^{-3}$ (power law)
 Scale-free: no characteristic degree scale
 Mean-field derivation predicts exponent = 3 exactly